Muhammad Rayyan Ahmed

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Cover Letter: Working Student in Opto Characterization

X-FAB Semiconductor Foundries Erfurt 2026-11-05

Dear Ms Heidolf,

I am applying for the Working Student position in Opto Characterization in Erfurt. Analysing and visualising measurement data with Python, building data-processing tools and exploring Generative AI for engineering workflows are things I already do, and I would like to apply them to sensor and semiconductor data.

I am an M.Sc. Data Science student at TU Ilmenau with a B.S. in Artificial Intelligence. I live and study in Ilmenau, so Erfurt is an easy commute.

- **Data analysis in Python,** I preprocess, analyse and visualise data with NumPy, Pandas and Matplotlib; in a computer vision project I built a classifier that reached 94.11% accuracy on a 968-image test set
- **Generative AI,** I built DocuMind, a RAG application with the Claude API that answers questions about technical PDF manuals with page citations, evaluated on a 22-question test set
- **Reports and documentation,** I documented data sources, calculations and results for every dashboard I built and presented results to business users and a client

I would analyse and visualise SPAD and CIS measurement data, help improve Python data-processing tools and try Generative AI for reporting and data handling. I have no background in electronics, LabVIEW or PCB setups, so I would learn the measurement side from your team and contribute the data and software side.

I can work up to 20 hours per week alongside my studies, starting immediately. I would be glad to discuss the role.

I work in English (IELTS 8.0) and am currently learning German (A1).